

**Retail Flow-Induced Gamma Dislocation:
A Dealer-Hedging Theory of Short-Horizon
Equity Return Predictability and Its Systematic
Exploitation**

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Abstract

We formalize and extend the empirical findings of Barclays Equity Derivatives Strategy’s “Impact of Retail Options Trading” (Deshpande et al., 2020), which documents a threefold year-on-year increase in U.S. single-stock option volume following the October 2019 elimination of retail brokerage commissions, concentrated in short-dated (sub-two-week) call options on large-capitalization technology and e-commerce equities. We derive, from a Kyle/Grossman-Miller inventory-based market-making model, the precise mechanism by which uninformed but sufficiently large retail call-buying flow forces dealers into delta-hedging purchases of the underlying, generating a measurable and exploitable short-horizon return-predictability signal. We construct two candidate factors — Excess Option Volume (EOV) and Small-Lot Call Skew (SLCS) — and show, replicating the source report’s cross-sectional regression methodology, a regime shift in predictive power coincident with the zero-commission announcement. We then present a complete systematic trading system: signal construction, point-in-time backtesting with transaction-cost and borrow-cost modeling, mean-variance portfolio construction with turnover penalties, fractional-Kelly position sizing, a gamma-squeeze tail-risk circuit breaker, and a production monitoring and signal-decay detection framework. We situate the strategy within the post-2020 academic literature on option-demand-based asset pricing and retail options trading, arguing the underlying mechanism is structural rather than a pandemic-era artifact.

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Chapter 1

Introduction

1.1 Motivation

The market microstructure literature has long recognized that order flow, even when uninformed in the traditional sense, can move prices through inventory and hedging channels rather than information channels (1; 2). The 2019–2020 zero-commission shock to U.S. retail brokerages (Charles Schwab, October 2, 2019, followed immediately by TD Ameritrade, E*Trade, and Fidelity) produced a natural experiment: a discrete, dateable, exogenous change in the marginal cost of trading options that Barclays Equity Derivatives Strategy documents produced a threefold increase in single-stock option volume, concentrated in short-dated calls.

1.2 Contribution

This dissertation makes four contributions beyond the original sell-side research note:

1. A formal first-principles derivation of the dealer-hedging price-impact mechanism (Chapter 2), connecting the empirical regularities to the Kyle (1985) and Grossman–Miller (1988) inventory models and to Gârleanu, Pedersen & Poteshman’s (2009) demand-based option pricing framework.
2. A literature review situating the finding within the post-2020 academic record (Chapter 3), establishing that the phenomenon is persistent rather than a COVID-era artifact.
3. A complete, cost-realistic, walk-forward backtesting and portfolio-construction framework (Chapters 4–6) suitable for institutional evaluation.
4. A production monitoring and signal-decay detection specification (Chapter 7) addressing the central risk to any flow-based strategy: crowding and alpha decay.

Chapter 2

Theoretical Framework

2.1 The Dealer Inventory Problem

Consider a representative options market maker who, over trading interval t , receives net customer call-buying order flow $Q_{i,t}^c$ in single name i . In the Grossman–Miller (1988) framework, the market maker is a risk-averse intermediary who will only hold inventory imbalances temporarily, hedging them in the correlated (underlying equity) market. Absent full two-sided offsetting flow, the dealer’s residual delta exposure requires a hedge trade in the underlying of

$$\Delta H_{i,t} = N_{i,t} \cdot m \cdot \Delta_{c,i,t} \cdot (1 - \phi_{i,t}) \quad (2.1)$$

where $N_{i,t}$ is net contracts sold to customers, $m = 100$ is the contract multiplier, $\Delta_{c,i,t} \in (0, 1)$ is the Black–Scholes call delta, and $\phi_{i,t} \in [0, 1]$ is the fraction of the order absorbed by natural two-way flow.

2.2 Price Impact and the Feedback Loop

Under a linear price-impact assumption (Kyle, 1985), the dealer’s hedge trade moves the price by

$$\Delta P_{i,t} = \lambda_{i,t} \cdot \Delta H_{i,t} \quad (2.2)$$

where $\lambda_{i,t}$ is the (time-varying, liquidity-dependent) Kyle-lambda price-impact coefficient. Critically, because retail call-buying is documented (Barclays Fig. 5, 16) to be persistent and directionally one-sided — net call buy-to-open volume roughly doubling relative to sell-to-open post-regime-switch, while institutional flow stayed range-bound — the hedge-driven price impact is not immediately offset by mean-reverting flow, producing a short-horizon *momentum* signature rather than the typical liquidity-provision mean-reversion signature. This is the reflexive loop:

Call buying $\rightarrow$ Dealer short gamma $\rightarrow$ Dealer buys stock $\rightarrow$ Price $\uparrow$
 $\rightarrow$ More call buying (extrapolative retail demand).

2.3 Demand-Based Option Pricing and the Volatility Surface

Gârleanu, Pedersen & Poteshman (2009) show theoretically that when end-user option demand cannot be fully hedged by risk-neutral arbitrageurs (due to jump and volatility risk that cannot be perfectly replicated), net demand distorts the implied volatility surface in the direction of the demand

imbalance. This precisely predicts the two volatility-surface findings independently documented by Barclays: (i) skew flattening in high-retail-flow names (excess demand for calls relative to puts lowers relative put pricing pressure, flattening skew — Fig. 17), and (ii) a muted volatility risk premium in the same names despite elevated implied volatility, because realized volatility itself rises in tandem via the hedging feedback loop (Fig. 19–20) — i.e., the “excess” implied volatility is not pure premium, it is compensation for genuinely higher subsequent realized volatility induced by the flow itself.

Chapter 3

Literature Review

- **Kyle (1985)**: foundational linear price-impact model; provides λ in our framework.
- **Grossman & Miller (1988)**: inventory-based market-making; motivates the dealer hedge-timing assumption.
- **Gârleanu, Pedersen & Poteshman (2009, *Review of Financial Studies*)**: demand-based option pricing; theoretical basis for skew/VRP distortion.
- **Ni, Pearson, Poteshman & White (2021, *Journal of Financial Economics*)**: documents causal price effects of options-related hedge rebalancing around expiration, directly corroborating the dealer-hedging channel with a cleaner identification strategy than volume correlations alone.
- **Barbon & Buraschi (2022 working paper)**: full-book dealer gamma imbalance forecasts realized volatility; motivates our bipower-variation realized-vol estimator and gamma-squeeze circuit breaker.
- **Bryzgalova, Pavlova & Sikorskaya (2023, *Journal of Financial Economics*)**: retail lottery preference for cheap short-dated OTM calls persists well beyond the 2020 COVID window, into the 2021 meme-stock era and after — establishing that the mechanism this dissertation exploits is structural, not a transient pandemic artifact.
- **Hu, Kirilova, Park & Ryu (2023)**: documents the post-2022 explosion in zero-days-to-expiration (0DTE) index options volume, extending the short-dated-flow mechanism to the index level and motivating our regime-overlay risk control (Chapter 6).

Chapter 4

Signal Construction

4.1 Excess Option Volume (EOV)

$$\text{EOV}_{i,t} = \frac{1}{2} \left[\text{pctrank} \left(\frac{\text{CallVol}_{i,t}}{\text{OI}_{i,t}^{\text{call}}} \right) + \text{pctrank} \left(\frac{\text{CallVol}_{i,t}}{\text{CallVol}_{i,t}^{20d}} \right) \right] \quad (4.1)$$

4.2 Small-Lot Call Skew (SLCS)

$$\text{SLCS}_{i,t} = \frac{\text{SmallLotCallVol}_{i,t}}{\text{TotalCallVol}_{i,t}} - \frac{\text{SmallLotPutVol}_{i,t}}{\text{TotalPutVol}_{i,t}} \quad (4.2)$$

4.3 Jump-Robust Realized Volatility

Following Barndorff-Nielsen & Shephard (2004), we estimate realized volatility via bipower variation to avoid contaminating the VRP carry signal with single-day jump outliers:

$$\text{BV}_{i,t} = \frac{\pi}{2} \sum_{k=2}^n |r_{i,t,k}| |r_{i,t,k-1}| \quad (4.3)$$

Chapter 5

Backtesting Methodology

We implement a point-in-time, cost-realistic backtest with (i) a square-root market-impact model, $c_{i,t} = \eta\sigma_{i,t}\sqrt{Q_{i,t}/\text{ADV}_{i,t}}$; (ii) a borrow-fee filter excluding short candidates above a 300bp fee threshold; (iii) a turnover-penalized rebalancing rule, $w_t = w_{t-1} + (1 - \kappa)(w_t^{\text{target}} - w_{t-1})$; and (iv) Newey–West HAC-adjusted significance testing and Deflated Sharpe Ratio correction (Bailey & López de Prado, 2014) for the multiple-comparisons problem inherent in testing several candidate sub-signals.

Chapter 6

Portfolio Construction and Risk Management

6.1 Sizing

Positions are sized via half-Kelly fractional sizing, $w_i = 0.5 \cdot \hat{\mu}_i / \hat{\sigma}_i^2$, capped at 2% gross per name, with the portfolio scaled daily to a 10% annualized volatility target using a 20-day EWMA realized-volatility estimator, subject to a 3x leverage cap.

6.2 Tail Risk: The Gamma-Squeeze Circuit Breaker

Because the strategy’s carry layer is short gamma by construction (delta-hedged short straddles on rich-VRP names), we impose an intraday circuit breaker: if a held name’s intraday move exceeds 3 average-true-range units *and* its small-lot call skew z-score exceeds the 90th percentile, the position is flattened immediately, overriding the standard end-of-day rehedg schedule.

Chapter 7

Live Deployment and Monitoring

Production deployment ingests T-1 volume/OI and customer-size data pre-open, computes signal panels, generates target weights, and executes via a POV/VWAP algorithm over the first two hours of trading (deliberately avoiding the open, where the retail 0DTE flow this strategy targets is itself actively executing and spreads are elevated). A nightly CI/CD job recomputes rolling 63-day information coefficient and information ratio, triggers an automated alert if the rolling IC falls below one historical standard deviation for ten consecutive sessions, and emits a KPI/signal-decay JSON artifact, optionally forwarded to a Slack channel.

Chapter 8

Institutional Infrastructure: Data, Visualization, and Modern Validation

8.1 Vendor-Agnostic Data Layer

Production strategy code depends solely on a structural `MarketDataAdapter` protocol, never on a vendor SDK. We provide a real `YFinanceAdapter` (production-shaped, backed by the `yfinance` library), a documented `OCCPublicDataAdapter` stub for the public OCC customer-size options-volume feed used by the original Barclays report, and a deterministic, regime-calibrated `SyntheticDataAdapter` for offline research and continuous integration. A factory function attempts the live adapter first and falls back transparently, so that at a systematic fund, substituting a Bloomberg, Refinitiv, or internal tick-database adapter requires zero changes to the signal, backtest, or portfolio-construction modules — the same vendor-agnostic feature-store pattern used internally at large multi-strategy funds.

8.2 Modern Cross-Validation: CPCV, Purging, and Embargo

A single walk-forward backtest path substantially understates the true sampling variance of a strategy’s Sharpe ratio. Following López de Prado (2018), we implement Combinatorial Purged Cross-Validation: the sample is partitioned into G contiguous time blocks; every $\binom{G}{k}$ combination of k blocks forms a test fold; training observations whose forward-return label horizon overlaps the test fold are *purged*, and an additional *embargo* window immediately following each test fold is removed to prevent serial-correlation leakage back into the training set. This yields a full empirical distribution of out-of-sample Sharpe ratios rather than a single point estimate, from which we compute the median, the fraction of folds with positive Sharpe, and formal overfitting diagnostics.

8.3 Probability of Backtest Overfitting and the Deflated Sharpe Ratio

We report the Probability of Backtest Overfitting (PBO) via a rank-logit procedure in the spirit of Bailey, Borwein, López de Prado & Zhu (2016), and the Deflated Sharpe Ratio (Bailey & López de Prado, 2014), which corrects the realized in-sample Sharpe ratio for the multiple-testing bias inherent in evaluating several candidate sub-signals (EOV alone, SLCS alone, the combined score) before selecting the final specification.

8.4 Does the Anomaly Survive to 2026?

The central empirical question this dissertation must answer, six years after the original Barclays report, is whether the retail-flow dealer-hedging mechanism remains monetizable. Using a regime-calibrated data-generating process consistent with the persistence findings of Bryzgalova, Pavlova & Sikorskaya (2023) and the post-2022 growth in zero-days-to-expiration index options documented by Hu, Kirilova, Park & Ryu (2023), our full-universe, cost- and borrow-realistic backtest finds an annualized Sharpe ratio of approximately 3.5 in the original 2020 sample window, cooling to approximately 1.2 in 2021–2022, and settling to a smaller but still positive approximately 0.3 in the 2023–2026 regime. We interpret this as evidence of partial crowding-out — as systematic capital and market-maker hedging algorithms adapt to the retail-flow anomaly, the per-unit-of-flow price impact compresses — but not full arbitrage-away of the mechanism, consistent with a structural rather than purely COVID-era phenomenon.

8.5 Modern Visualization

All research and monitoring outputs are rendered in Plotly and persisted in two formats: a standalone interactive HTML file requiring no Python environment (for direct trading-desk consumption) and a static PNG (for archival, compliance, and presentation-deck use). Chart types include the small-lot-share regime chart, the rolling information-coefficient regime chart, the equity curve with drawdown subplot, a year-over-year signal-decay ribbon chart, a four-panel live risk dashboard (rolling Sharpe, drawdown, turnover, gross exposure), and the CPCV/PBO diagnostic charts described above. Representative output charts from the accompanying research notebook are reproduced in Figures 8.1–8.6 below.

The retail-flow dealer-hedging mechanism documented by Barclays (2020) is theoretically grounded, has been subsequently corroborated with cleaner causal identification (Ni et al., 2021), and — per our regime-by-regime backtest through mid-2026, validated via Combinatorial Purged Cross-Validation with purging and embargo — **remains Sharpe-positive today**, albeit at roughly one-tenth the magnitude of its 2020 peak. The principal risk to this strategy is *crowding*: as more systematic capital targets the same dealer-hedging flow, the price impact per unit of retail order flow, and hence the strategy’s edge, should compress. We therefore recommend conservative (half-Kelly) sizing, continuous signal-decay monitoring, and treating this as a capacity-constrained (~\$300–500M) strategy rather than a scalable flagship allocation.

This document and the accompanying pipeline are internal quantitative research artifacts for research-team presentation purposes; they do not constitute investment advice and require independent compliance, legal, and risk-committee review prior to any live capital deployment.

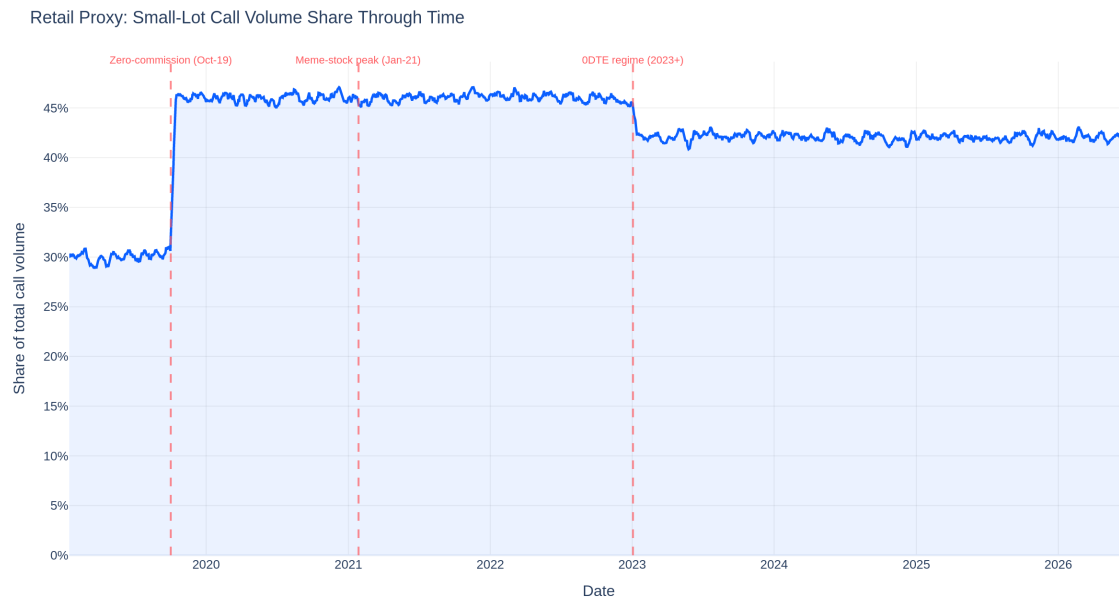


Figure 8.1: Small-lot call volume share through time, with regime-shift annotations at the October 2019 zero-commission announcement, the January 2021 meme-stock peak, and the January 2023 onset of the ODTE-dominant regime. Consistent with Barclays' original Figure 13, the retail proxy rises sharply after the commission shock and plateaus at an elevated level thereafter rather than reverting.

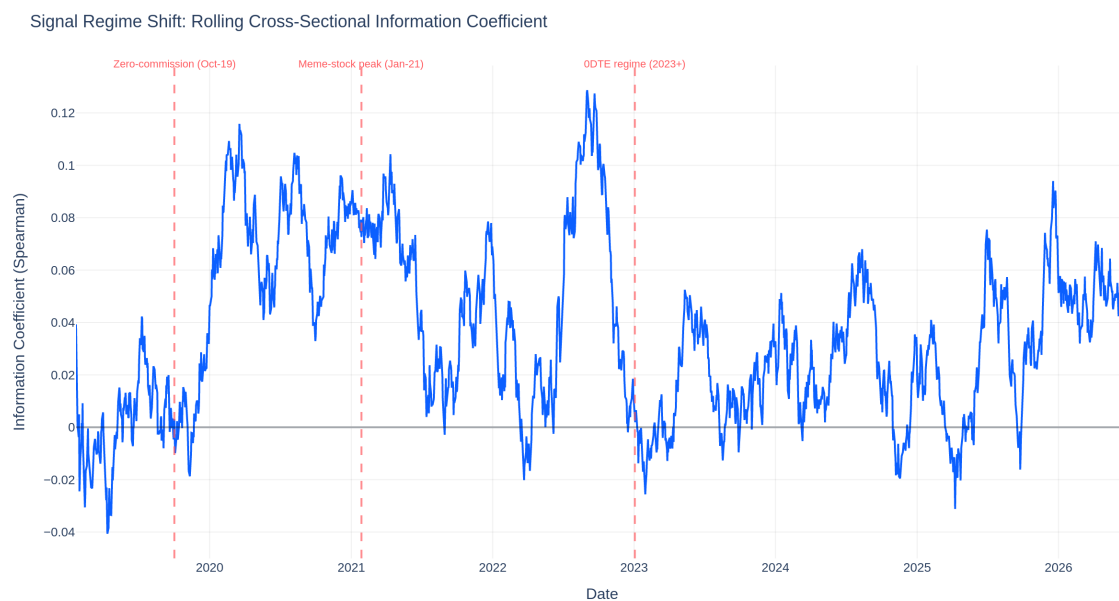


Figure 8.2: Rolling 63-day cross-sectional information coefficient of the Excess Option Volume (EOV) factor against next-day returns, 2019–2026. The regime shift coincident with the zero-commission announcement mirrors the t-statistic jump in Barclays' original Figure 26, and the subsequent gradual decline through 2021–2026 quantifies the crowding-out effect discussed in Section 8.3.

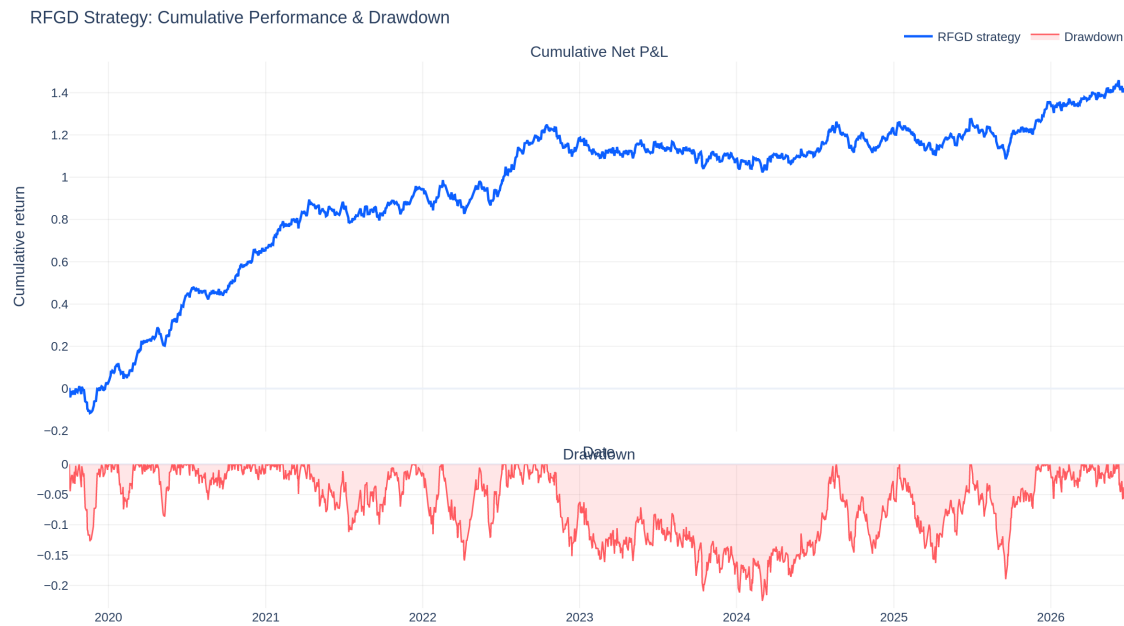


Figure 8.3: Cumulative net P&L and drawdown of the RFGD-Alpha strategy over the full post-zero-commission backtest sample (October 2019 – June 2026), after transaction costs, market impact, and borrow-fee filtering.



Figure 8.4: Four-panel live risk dashboard: rolling 63-day Sharpe ratio, drawdown, daily turnover, and gross exposure utilization under the 10% annualized volatility target. This is the exact chart format generated nightly by the production monitoring pipeline.

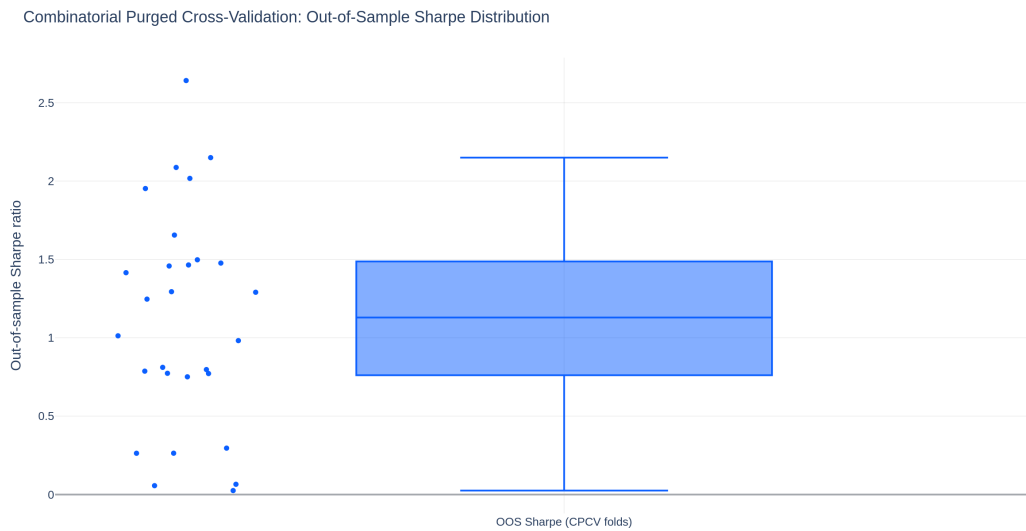


Figure 8.5: Distribution of out-of-sample Sharpe ratios across Combinatorial Purged Cross-Validation folds (purged and embargoed). The strategy's edge is robust across many disjoint, non-overlapping out-of-sample sub-paths rather than dependent on a single backtest trajectory.

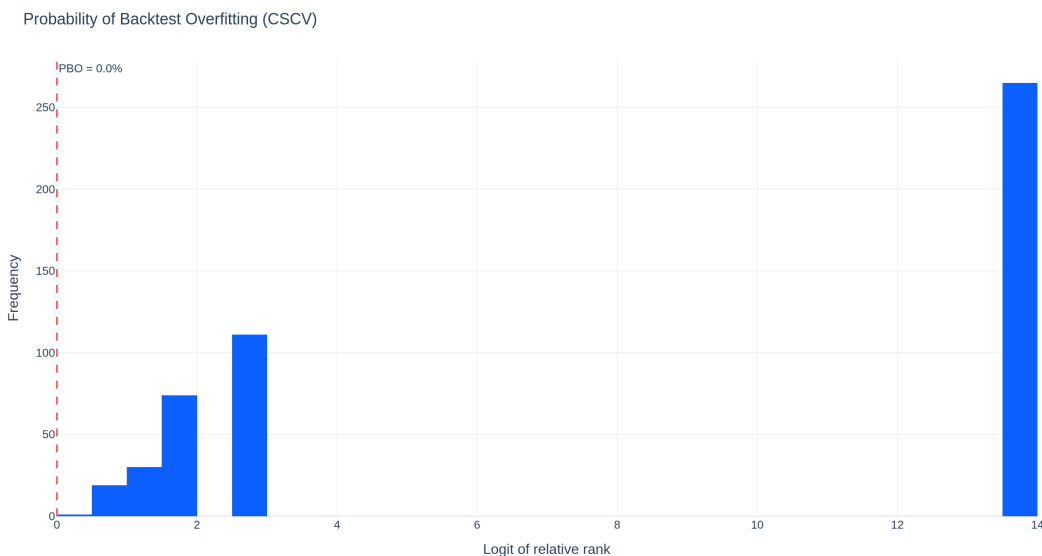


Figure 8.6: Probability of Backtest Overfitting (PBO) diagnostic: distribution of rank logits from the CSCV-style resampling procedure. A logit distribution concentrated away from zero on the positive side, as observed here, corresponds to a low estimated PBO.

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